

Local Nonlinear Projection-Based Reduced-Order Models

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September 10, 2026

Problem: 2D Burgers, mesh 250×250 .

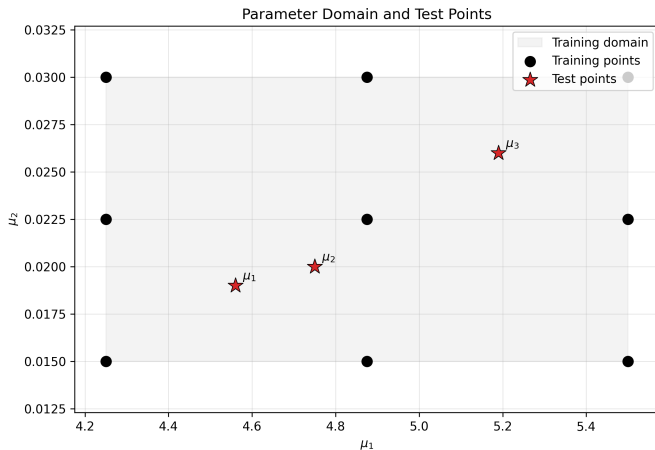
Test points:

$$\mu_1 = (4.56, 0.019), \quad \mu_2 = (4.75, 0.020), \quad \mu_3 = (5.19, 0.026).$$

Methods included:

- ▶ Global HPROM (linear POD)
- ▶ Global HQPROM (quadratic manifold)
- ▶ Global HPROM-GPR
- ▶ Local HPROM
- ▶ Local HQPROM
- ▶ Local HPROM-GPR

Parameter Domain and Test Points



Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg \min_{\mathbf{q}} \frac{1}{2} \|\mathbf{r}(\mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}))\|_2^2.$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta \mathbf{q}$ is obtained from

$$(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}) \Delta \mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}}, \quad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta \mathbf{q}.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad n = 96.$$

Hence

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \quad \boldsymbol{\psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}), \quad n = 39.$$

Thus

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}(\mathbf{q} \otimes \mathbf{q})].$$

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \quad n = 20, \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}),$$

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}} = \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters $k = 1, \dots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\text{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\text{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\text{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg \min_{\ell=1,\dots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \quad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^+ = \arg \min_{\ell=1,\dots,N_c} S_{k^-, \ell}^{\text{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{lin}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + d_{k^-, \ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{quad}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{quad}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) + d_{k^-, \ell},$$

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}).$$

HPROM-GPR selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \bar{\mathbf{g}}_{k^-, \ell}^T \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}) + d_{k^-, \ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}_{k^-}), \quad \bar{\mathbf{g}}_{k^-, \ell} := \bar{\mathbf{V}}_{k^-}^T (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$k^+ = k^- \Rightarrow$ no transfer, $k^+ \neq k^- \Rightarrow$ solve transfer problem.

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\text{lin}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+} \mathbf{q}) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\text{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\text{nl}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\text{nl}}(\mathbf{q}) \right\|_2^2.$$

Cluster Change: Nonlinear Least Squares

For nonlinear local manifolds, define

$$\mathbf{r}_{\text{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \quad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}) \Delta \mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\text{tr}}(\mathbf{q}^{(j)}), \quad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta \mathbf{q}.$$

Online Performance (250×250): Avg/Max Error over 3 Test Points

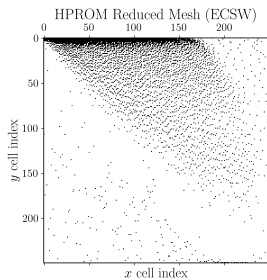
Computational model	n (N for HDM)	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,u}$ (%)	max $\mathbb{RE}_{2,u}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	–	12.517	–
HPROM	96	–	4 241	1.030	1.096	0.366	34.2
HQPPROM	39	–	3 460	0.818	0.861	0.641	19.5
HPROM-GPR	20	131	1 799	0.666	0.785	0.236	53.1
Local HPROM ($N_c = 3$)	35–48	–	3 405	0.897	0.933	0.154	81.4
Local HQPPROM ($N_c = 3$)	21–27	–	2 152	0.888	1.025	0.215	58.3
Local HPROM-GPR ($N_c = 3$)	9	60–97	811	0.714	0.786	0.113	110.5
Local HPROM ($N_c = 5$)	23–31	–	2 384	0.915	0.943	0.095	132.0
Local HQPPROM ($N_c = 5$)	15–20	–	1 593	0.760	0.878	0.128	97.9
Local HPROM-GPR ($N_c = 5$)	7	41–68	631	0.673	0.880	0.094	133.2
Local HPROM ($N_c = 8$)	16–26	–	1 781	0.787	0.900	0.070	179.7
Local HQPPROM ($N_c = 8$)	10–15	–	1 075	0.787	0.890	0.082	151.9
Local HPROM-GPR ($N_c = 8$)	6	28–54	541	0.801	1.156	0.085	147.9

Speedup is computed with respect to the HDM runtime, $t_{\text{HDM}} = 751.043$ s, averaged over the three test points.

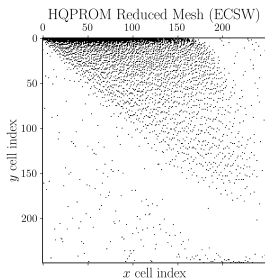
N_e denotes the number of sampled ECSW elements ($N_e = 62,500$ for HDM/full mesh).

All models are reported at a matched accuracy target of $1\% \pm 0.2\%$ in $\max \mathbb{RE}_{2,u}$: for each N_c the accuracy knob is tuned to that window (pod_tol for Local HPROM, ζ_{qua} for Local HQPPROM, n for Local HPROM-GPR), so differences in n , N_e and speedup are read at constant accuracy.

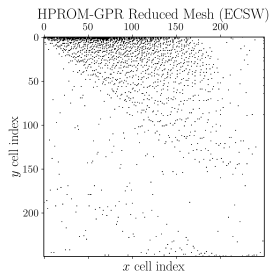
Global Reduced Meshes (ECSW)



HPROM ($N_e = 4,241$)

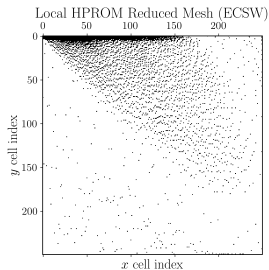


HQPROM ($N_e = 3,460$)

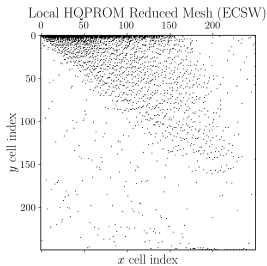


HPROM-GPR ($N_e = 1,799$)

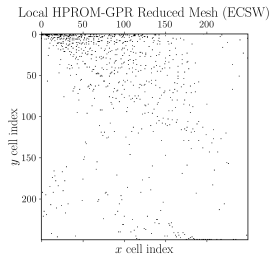
Local Reduced Meshes (ECSW), $N_c = 3$



Local HPROM ($N_e = 3,405$)

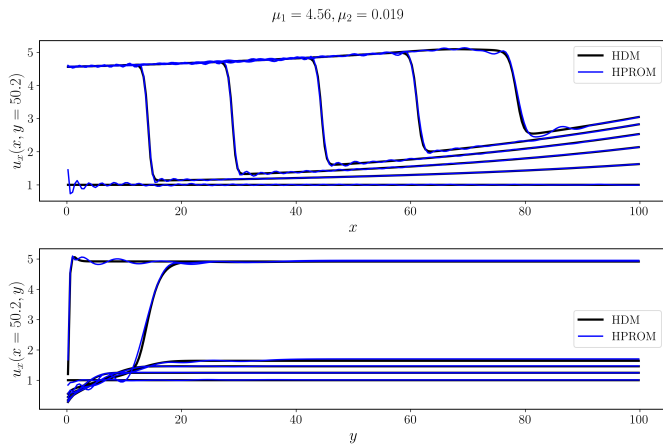


Local HQPROM ($N_e = 2,152$)

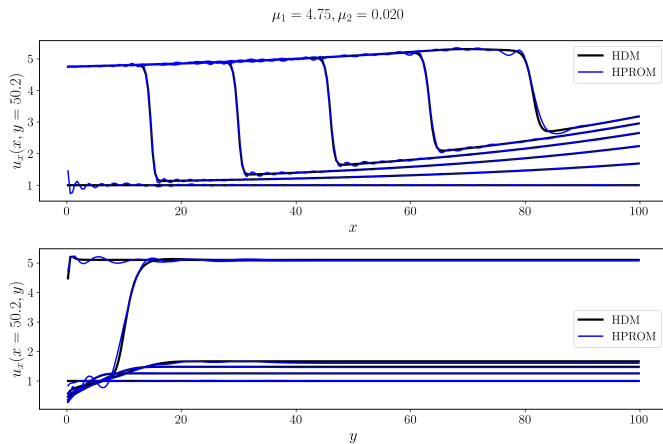


Local HPROM-GPR ($N_e = 811$)

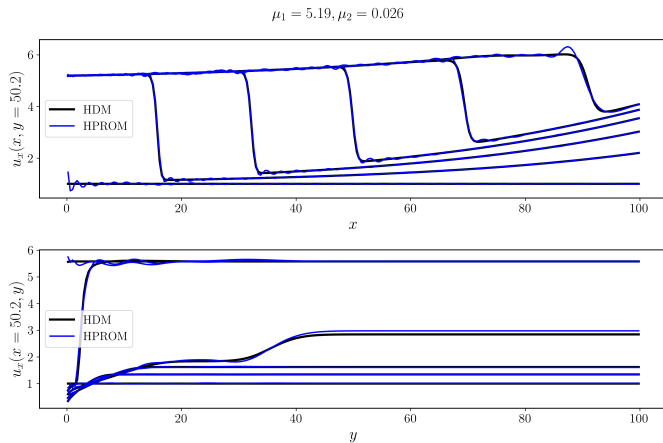
HPROM vs HDM: $\mu = (4.56, 0.019)$



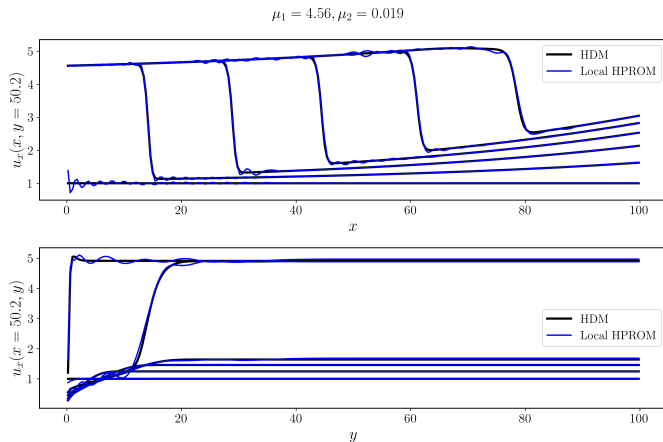
HPROM vs HDM: $\mu = (4.75, 0.020)$



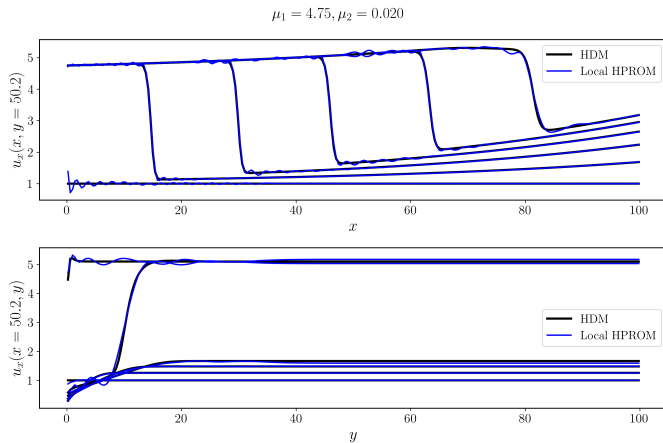
HPROM vs HDM: $\mu = (5.19, 0.026)$



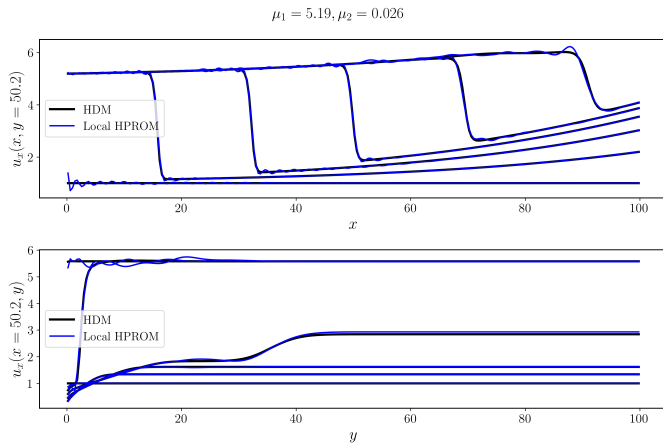
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$



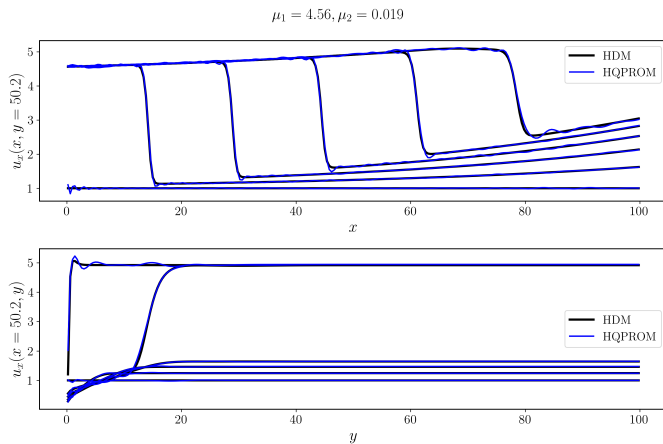
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$



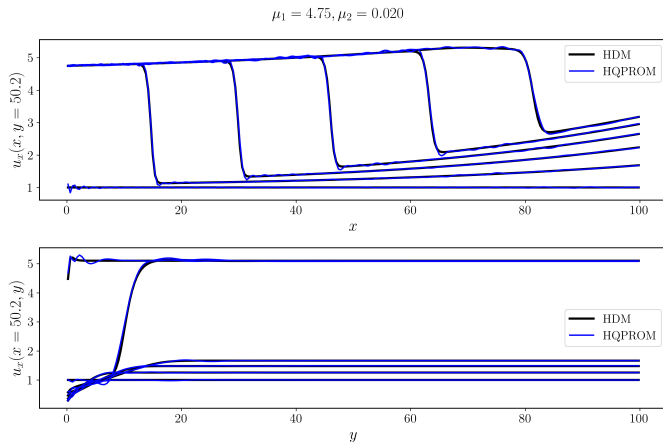
Local HPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$



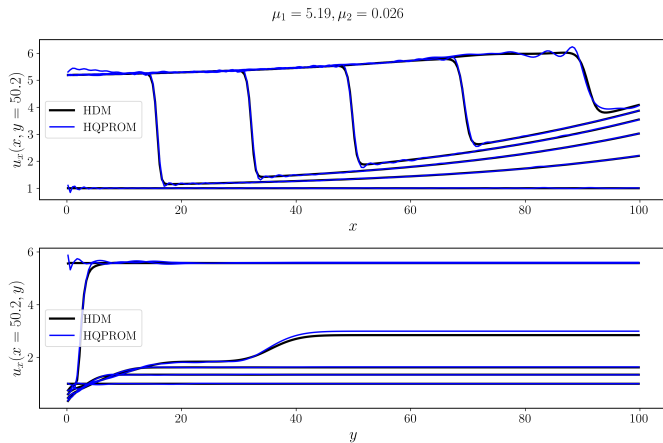
HQPROM vs HDM: $\mu = (4.56, 0.019)$



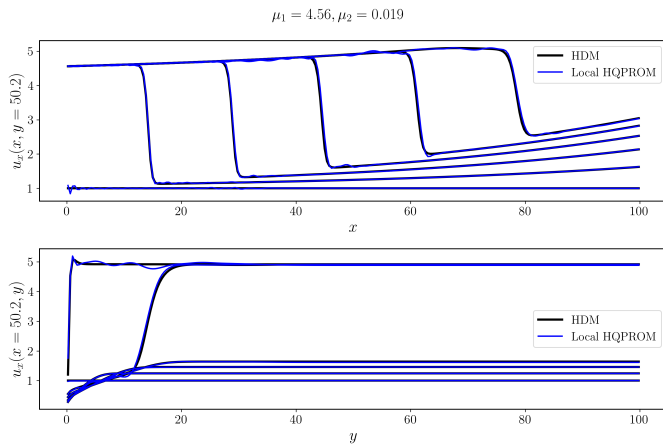
HQPROM vs HDM: $\mu = (4.75, 0.020)$



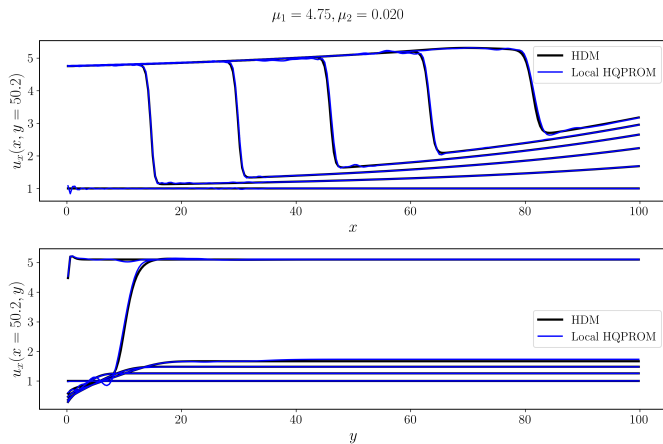
HQPROM vs HDM: $\mu = (5.19, 0.026)$



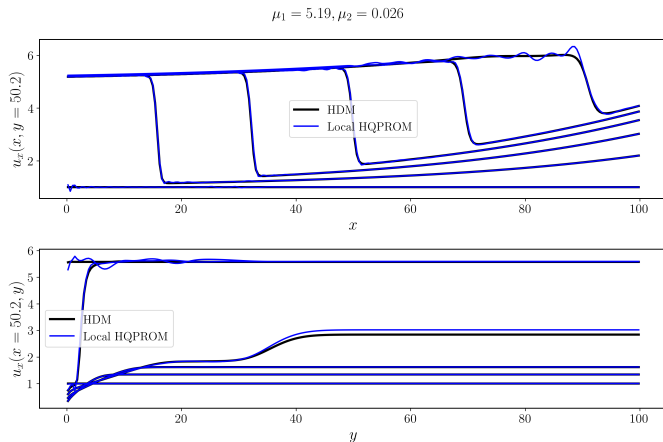
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$



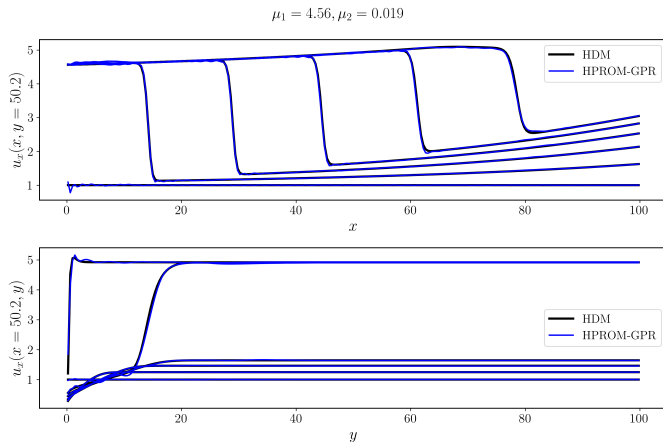
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$



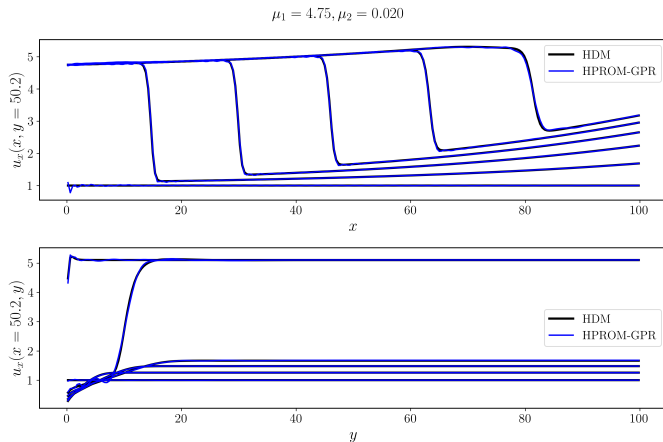
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$



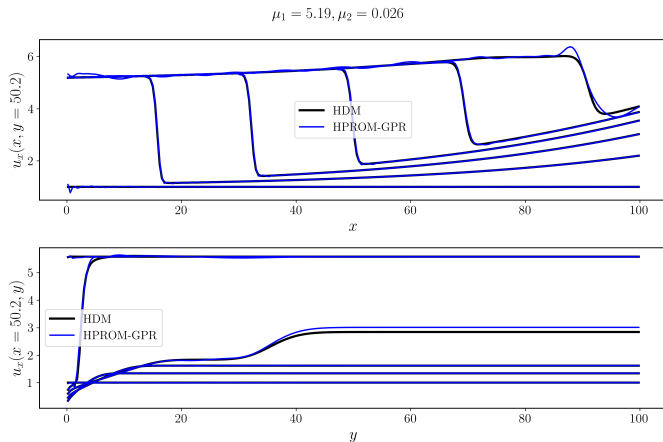
HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$



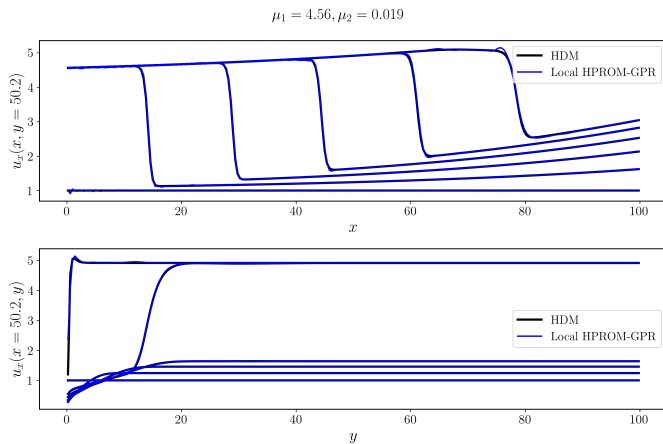
HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$



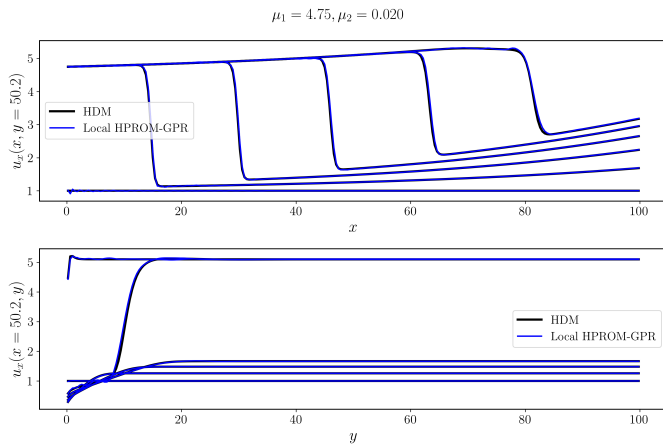
HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$



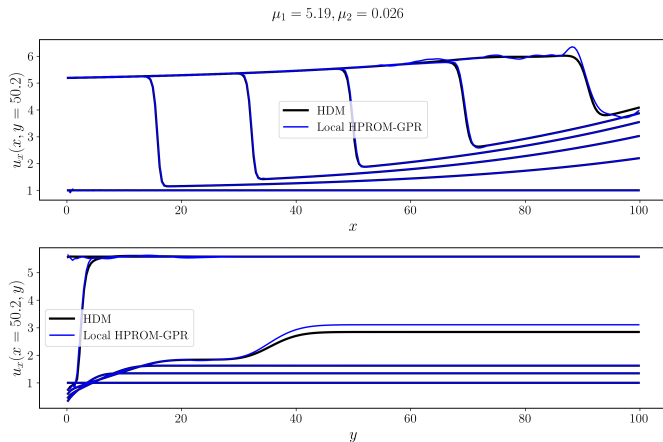
Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$



Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$



Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$



Appendix: Linear Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-} \mathbf{q}_{k^-}, \quad k^+ = \arg \min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

$$\text{Let } \mathbf{a}_{\ell} := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-} \mathbf{q}_{k^-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_{\ell} + \mathbf{b}\|_2^2 = \|\mathbf{a}_{\ell}\|_2^2 + \|\mathbf{b}\|_2^2 + 2\mathbf{a}_{\ell}^T \mathbf{b}.$$

$$\|\mathbf{b}\|_2^2 = \|\mathbf{V}_{k^-} \mathbf{q}_{k^-}\|_2^2 \text{ is independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-} \mathbf{q}_{k^-} \right].$$

Precompute

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2,$$

which yields

$$k^+ = \arg \min_{\ell} \left(2\mathbf{g}_{k^-, \ell}^T \mathbf{q}_{k^-} + d_{k^-, \ell} \right).$$

Appendix: HQPROM Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-} \mathbf{q}_{k^-} + \mathbf{H}_{k^-} (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-} \mathbf{q}_{k^-}, \quad \mathbf{c} := \mathbf{H}_{k^-} (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}).$$

$$k^+ = \arg \min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \arg \min_{\ell} \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2.$$

$$\|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\mathbf{a}_\ell^T \mathbf{b} + 2\mathbf{a}_\ell^T \mathbf{c} + 2\mathbf{b}^T \mathbf{c}.$$

$$\|\mathbf{b}\|_2^2, \|\mathbf{c}\|_2^2, 2\mathbf{b}^T \mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 \right.$$

$$\left. + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-} \mathbf{q}_{k^-} + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k^-} (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}) \right].$$

Define

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}),$$

then

$$k^+ = \arg \min_{\ell} \left(2\mathbf{g}_{k^-, \ell}^T \mathbf{q}_{k^-} + 2\mathbf{m}_{k^-, \ell}^T (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}) + d_{k^-, \ell} \right).$$

Appendix: HPRM-GPR Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-} \mathbf{q}_{k-} + \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-}, \quad \bar{\mathbf{q}}_{k-} = \mathcal{N}_{\text{GPR},k-}(\mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-} \mathbf{q}_{k-}, \quad \mathbf{c} := \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2$$

$$= \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\mathbf{a}_\ell^T \mathbf{b} + 2\mathbf{a}_\ell^T \mathbf{c} + 2\mathbf{b}^T \mathbf{c}.$$

$$\|\mathbf{b}\|_2^2, \|\mathbf{c}\|_2^2, 2\mathbf{b}^T \mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k-} \mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-} \right].$$

Hence, with

$$\bar{\mathbf{g}}_{k-, \ell} := \bar{\mathbf{V}}_{k-}^T (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}),$$

$$k^+ = \arg \min_{\ell} \left(2\mathbf{g}_{k-, \ell}^T \mathbf{q}_{k-} + 2\bar{\mathbf{g}}_{k-, \ell}^T \bar{\mathbf{q}}_{k-} + d_{k-, \ell} \right).$$